

Harnessing the Power of Knowledge Graphs to Improve Causal Discovery

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Abstract—Reconstructing the structure of causal graphical models from observational data is crucial for identifying causal mechanisms in scientific research. However, real-world noise and hidden factors can make it difficult to detect true underlying causal relationships. Current methods mainly rely on extensive expert analysis to correct wrongly identified connections, guiding structure learning toward more accurate causal interactions. This reliance on expert input demands significant manual effort and is risky due to potential erroneous judgments when handling complex causal interactions. To address these issues, this paper introduces a new, expert-free method to improve causal discovery. By utilizing the extensive resources of static knowledge bases across various fields, specifically knowledge graphs (KGs), we extract causal information related to the variables of interest and use these as prior constraints in the structure learning process. Unlike the detailed constraints provided by expert analysis, the information from KGs is more general, indicating the presence of certain paths without specifying exact connections or their lengths. We incorporate these constraints in a soft way to reduce potential noise in the KG-derived priors, ensuring that our method remains reliable. Moreover, we provide interfaces for various mainstream causal discovery methods to enhance the utility of our approach. For empirical validation, we apply our approach across multiple areas of causal discovery. The results show that our method effectively enhances data-based causal discovery and demonstrates its promising applications.

Index Terms—Causal discovery, structure learning, Bayesian network, knowledge graph.

I. INTRODUCTION

MODELING causal relationships among variables using directed acyclic graphs (DAGs) is a fundamental approach for causal representation [1]. Recovering the structure of these DAGs from observational data is crucial for uncovering new causal mechanisms in science [2], [3], [4], [5], [6], [7], [8], such as building regulatory networks [9], studying brain activity [10], and constructing econometric models [11]. The

identified causal mechanisms can provide novel insights across various domains and support more detailed causal analyses, such as the quantitative estimation of causal effects [12], [13].

However, accurately recovering the true causal structure is challenging due to practical limitations, including data insufficiency, noise, and interference from unobserved factors [14], [15], [16], [17]. To enhance the quality of recovered causality, a common approach involves relying on expert analysis to derive prior constraints for structure learning [18], particularly for correcting the presence of edges [19], [20]. These edge-level constraints guide the structure learning process toward more meaningful and accurate causal relationships. However, expert-driven causal discovery requires considerable manual effort to analyze the causal interactions among variables and is vulnerable to inaccuracies from expert judgments [21], especially when causal interactions extend beyond existing knowledge.

To address these challenges, this paper proposes a novel expert-free approach to mitigate data limitations by extracting prior causal knowledge from domain-specific knowledge bases to inform causal discovery. Recent advancements in knowledge engineering have enabled the organization of extensive domain knowledge into structured repositories known as knowledge graphs (KGs) across various fields [22], [23], [24], [25]. These KGs serve as rich resources for deriving prior causal information, which are of great potential to enhance the causal discovery process. However, the causal knowledge derived from KGs is relatively abstract and requires careful integration into structure learning to be effective.

For incorporating KG-derived causal knowledge, we employ ancestral constraints, which assert the existence of a directed path from a cause to an effect variable. This approach is motivated by the static nature of KG-derived causal information, which typically represents qualitative causality without detailing the fine-grained interactions among variables or specifying direct causal edges. In practice, qualitative causality in KGs is often captured indirectly through multiple causal interactions, forming directed paths composed of several edges. Therefore, we utilize the existence of directed paths to characterize prior constraints from KGs rather than relying on precise edge-level constraints, aligning our method with the inherent ambiguity and generality of KG-derived knowledge.

To integrate these KG-driven ancestral constraints into the structure learning process, we adopt a soft integration method that accommodates potential inaccuracies in the prior knowledge. Specifically, we extend the scoring function of the causal structure by incorporating a bonus term that rewards the

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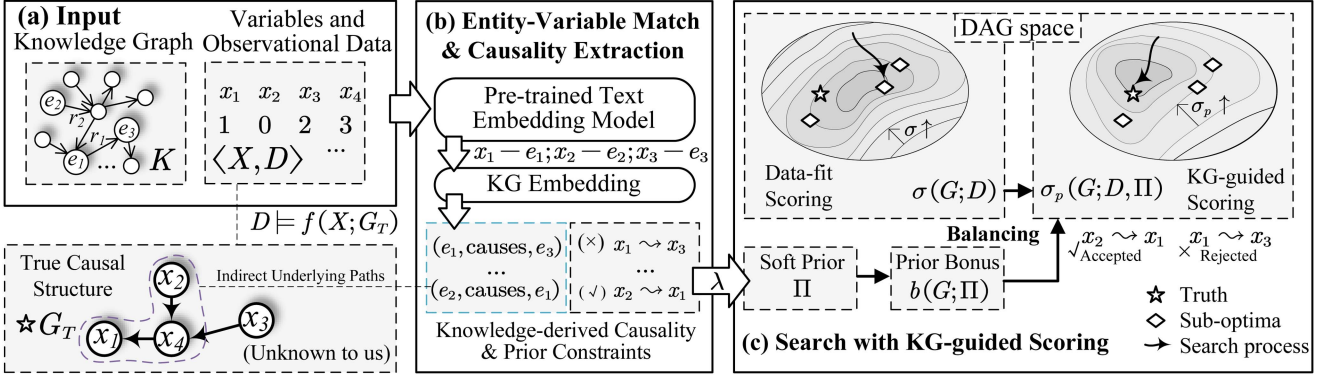


Fig. 1. The diagram of the proposed KG-driven CSL framework. (a) The framework begins by taking an external knowledge graph K , descriptions of variables X , and observational data D as inputs. (b) Subsequently, prior causal knowledge is extracted from the KG by entity-variable match and causal triplet extraction. (c) KG-derived prior knowledge is used as ancestral constraints by softly extending the data approximation score, reshaping the score distribution of the DAG space and guiding structure learning to the genuine underlying causality.

satisfaction of KG-derived prior constraints. The weight of this prior-driven term is adjustable, allowing modulation of the priors' influence on the overall structure learning. Constraints that do not sufficiently compensate for their associated penalties through adherence bonuses are discarded, thereby preventing erroneous priors from significantly conflicting with data-driven approximations. This soft integration approach ensures the robustness of our method against potential noise and inaccuracies in the KG-derived priors.

Furthermore, we provide strategies for incorporating KG-derived priors into various mainstream structure learning methodologies, including constraint-based and gradient-based approaches. This adaptability enhances the utility and versatility of our framework, enabling its effective application across different causal discovery algorithms and research contexts.

For evaluation, we utilize various KGs, including both small-scale domain-specific KGs and large general-purpose KGs, as sources of prior causal knowledge. We test our KG-driven causal discovery framework on datasets from diverse domains with varying numbers of variables. The results demonstrate a significant improvement in data-driven causal discovery performance and highlight the promising practical potential of our approach for domain-specific causal analysis. Fig. 1 provides a visual illustration of the KG-guided causal structure learning (CSL) framework. The main contributions of this paper are summarized as follows:

- To the best of our knowledge, this is the first study to propose a generalized, expert-free, KG-driven framework to enhance causal discovery.
- We introduce a soft ancestral constraint-based structure learning framework that aligns prior constraints with the relatively weak knowledge derived from KGs and incorporates tolerance for prior errors.
- We develop specific strategies to adapt our method to different structure learning algorithms, enhancing its utility. Empirical evidence demonstrates the practical potential of our approach.

The rest of this paper is organized as follows. Section II introduces the preliminaries and related work. Sections III-A and III-B describe the proposed framework. Section IV presents the empirical evaluation of our framework, and Section V concludes the paper.

II. BACKGROUND

This section introduces the essential concepts of causal structure learning and reviews closely related work pertinent to this study.

A. Causal Structure Learning

Causal structure learning (CSL) aims to reconstruct a causal Directed Acyclic Graph (DAG) from observational data or a combination of observational and interventional data [26], [27]. Observational data consists of independent and identically distributed (i.i.d.) samples generated from the true causal graphical model, while interventional data is obtained from controlled experiments that help identify parts of the causal structure that cannot be uniquely determined from observational data alone.

In this paper, we assume the use of discrete Bayesian Networks (BNs) [28] as the underlying causal graphical models and operate under the assumption that there are no hidden variables [29]. Furthermore, our objective is to enhance causal structure learning using only observational data, without the necessity for additional interventional experiments. Consequently, the subsequent discussions focus exclusively on CSL from observational data.

The joint distribution of variables encoded by a BN can be expressed using the Markov property as follows:

$$P(X_1, X_2, \dots, X_n) = \prod_{i=1}^n P(X_i | \mathbf{Pa}_i^G), \quad (1)$$

where $X_1, X_2, \dots, X_n$ are the variables of interest, G represents the BN, and $\mathbf{Pa}_i^G$ denotes the set of parent nodes of X_i in the BN.

Leveraging this property, the likelihood function of a BN G given a set of observational data $D = \{d_1, d_2, \dots, d_m\}$, where D consists of m i.i.d. samples generated based on the true BN, is formulated as:

$$L(G | D) = P(D | G) = \prod_{i=1}^m P(d_i | G) \\ = \prod_{i=1}^m \prod_{j=1}^n P(X_j = x_{ij} | \mathbf{Pa}_j^G = pa_{ij}), \quad (2)$$

where x_{ij} represents the value of variable X_j in data point d_i , and pa_{ij} denotes the values of the parents of X_j in G as observed in data point d_i .

Based on this likelihood, score-based methods are employed for CSL tasks. These methods involve defining a scoring criterion for BNs to assess their fit to the data and subsequently searching for the optimal DAG that maximizes this score. Score-based methods are particularly emphasized in this paper due to their flexibility and extensibility in incorporating prior constraints. For a comprehensive overview of alternative CSL approaches, including constraint-based and hybrid methods, readers are referred to the survey by Scanagatta et al. [30].

Well-known scoring functions for Bayesian Networks (BNs) compute the logarithm of the likelihood function defined in (2) under specific assumptions.¹ These functions then incorporate a penalty for model complexity to formalize the scoring criteria of BNs. Popular scoring functions include the Bayesian Information Criterion (BIC) [31], Bayesian Dirichlet equivalent uniform (BDeu) [32], and Minimum Description Length (MDL) [33] scores, among others.

Typically, these scores are decomposable, meaning they can be computed locally for each variable and its parents and then summed to obtain the total score:

$$\sigma(G | D) = \sum_{i=1}^n \sigma(X_i, \mathbf{Pa}_i^G; D) \quad (3)$$

This property allows the search problem to be broken down into n local optimization tasks, subject to the acyclicity constraint of the DAG. Well-known search algorithms employed for structure learning include Greedy Equivalence Search (GES) [34], Hill Climbing (HC) [32], and Tabu Search [35].

B. Causality and Knowledge Graphs

Knowledge graphs (KGs) organize information structurally in the form of triplets, (h, r, t) , where h and t represent the head and tail entities, respectively, and r denotes the relation from the head to the tail entity [36]. This structural representation is similar to graphical models that encode causality. However, unlike causal graphical models, KGs originate from human knowledge rather than data analysis and can represent a wide variety of relations beyond causality.

With the growing demand for explainability in AI applications [37], causality has become increasingly important in

knowledge engineering. Numerous techniques have been developed to extract structural causal knowledge from both unstructured data [38], [39], [40], [41], [42], such as natural language texts, and structured data [43], including existing knowledge graphs or databases. These causal knowledge graphs span various significant domains, including clinical decision-making [44], healthcare [23], medical science [43], autonomous driving [45], social science [46], and economics [47], among others. However, related studies primarily belong to the knowledge engineering domain and focus on extracting and organizing causal knowledge rather than recovering causality from data. In contrast, this paper employs knowledge-level causality as qualitative priors to guide data-based causal discovery. This approach represents a practical application of knowledge engineering within the domain of causal learning.

C. Connection to LLM-Driven Causal Discovery

Recently, a series of studies have proposed Large Language Model (LLM)-driven causal discovery [48], [49], [50], which utilize LLMs as causal experts to provide insights for structure learning. These studies share a similar objective with our work by aiming to construct an expert-free framework to enhance causal discovery, leveraging the strong generalization capabilities of LLMs to supply priors across various domains.

However, LLMs currently face significant challenges in accurately identifying causal relationships [51], [52]. They are prone to generating misleading or incorrect causal inferences, which can adversely affect the structure learning process, as evidenced in the study [53]. Additionally, LLMs can suffer from consistency issues, often producing different responses to identical causal queries, thereby undermining the stability of the causal discovery process. In contrast, knowledge stored in static knowledge bases (KGs) tends to be more stable and trustworthy, as supported by our empirical results.

Therefore, KG-driven causal discovery offers greater reliability compared to LLM-based approaches at the current stage of development. Moreover, the prior constraint-based methodology proposed in this paper is adaptable to LLM-driven approaches by simply substituting KG-extracted priors with those derived from LLMs. This flexibility allows our framework to benefit from advancements in LLM-driven causal discovery while maintaining the robustness provided by structured knowledge bases.

III. KNOWLEDGE GRAPH-DRIVEN CAUSAL DISCOVERY

This section details the proposed Knowledge Graph (KG)-driven causal discovery framework. To set the stage, we first define the key symbols used in our approach.

The inputs to the framework consist of a set of variables $X = \{X_i\}_{i=1}^d$, each accompanied by textual descriptions. Additionally, we have a dataset $D \in \mathbb{N}^{m \times d}$, comprising m samples generated based on the underlying DAG $G_T(X, E(G_T))$ of these variables, where $E(G_T) \subseteq X \times X$ represents the set of edges in G_T . Furthermore, a knowledge graph $K(E, R, T)$ is provided, which includes a set of entities $E = \{e_i\}_{i=1}^N$, a relation set R containing a subset of causal relationships $R_c = \{r_{c_i}\}_{i=1}^q \subseteq R$, and a set of triplets $T \subseteq E \times R \times E$. The

¹Some scoring functions do not directly compute the logarithm of the likelihood as in (2). However, they generally follow the same decomposable principle, allowing local calculations that are subsequently aggregated.

objective of our framework is to recover the true causal structure G_T from the observational data D , leveraging the prior causal knowledge embedded within the KG K .

A. Causal Knowledge Derivation From the KG

This section details the process of extracting causal knowledge related to the variable set X from the knowledge graph K . The procedure involves two main steps: identifying entities that correspond to each variable using text embeddings and extracting triplets with causal relations among these entities from T . The process is elaborated below.

Matching Data Variables to KG Entities: For brevity, we use the same symbol X_i to represent the descriptive text of the corresponding variable. To identify entities in E that match the variables, we calculate the embeddings of both variable descriptions and entities using pre-trained models. Entities that describe the same concept as variable X_i are identified based on embedding similarity. Specifically, the matching process is defined as:

$$M(X_i) = \left\{ e_j \mid \frac{h^{e_j} \cdot h^{X_i}}{\|h^{e_j}\| \|h^{X_i}\|} > \alpha, e_j \in E \right\} \quad (4)$$

where h^{X_i} represents the embedding of variable X_i and h^{e_j} is the embedding of entity e_j . The threshold $\alpha \in (0, 1)$ is preset to determine the similarity required for a variable-to-entity match. The set of entities corresponding to any variable in X is denoted as $E_V = \bigcup_{X_i \in X} M(X_i)$. Causal relations among these entities are subsequently extracted to derive prior knowledge.

Remark 1: A single variable may correspond to multiple entities. For example, the variable “age” can be represented by entities such as “old” and “young,” which describe distinct values or states of “age.”

Extracting Causal Knowledge from the KG: To extract prior knowledge, we identify triplets with causal relations $r \in R_c$ and entities E_V that correspond to specific variables. We employ two strategies based on the scale of the KG to balance efficiency and knowledge quality.

For relatively small-scale KGs, we perform KG embedding and complete the triples with causal relations to extract more comprehensive causal knowledge. We utilize the TransE model [54], which defines the following scoring function:

$$f(\mathbf{h}, \mathbf{r}, \mathbf{t}) = -\|\mathbf{h} + \mathbf{r} - \mathbf{t}\|_{\ell_{1/2}} \quad (5)$$

where $\mathbf{h}$ and $\mathbf{t}$ are the embeddings of the head and tail entities, respectively, and $\mathbf{r}$ is the embedding of the relation in a triplet. The norm $\|\cdot\|_{\ell_{1/2}}$ represents the fractional norm.

All entities and relations are embedded into vectors by minimizing the score of triplets contained in T and maximizing the score of those not in T :

$$\mathcal{L} = \sum_{(h,r,t) \in T} \sum_{(h',r',t') \notin T} [f(\mathbf{h}, \mathbf{r}, \mathbf{t}) - f(\mathbf{h}', \mathbf{r}', \mathbf{t}')] \quad (6)$$

Here, the bold symbols represent the embedded vectors of the corresponding entities or relations. For example, $\mathbf{h}$ is the embedding of entity $h \in E$. The scoring function $f(\cdot)$ is defined in (5).

With the embeddings of entities and relations, causal knowledge is extracted as follows:

$$S = \{(h, r, t) \mid h, t \in E_V, r \in R_c, f(\mathbf{h}, \mathbf{r}, \mathbf{t}) < \beta\} \quad (7)$$

where β is the threshold for identifying the existence of a triplet, defined as:

$$\beta = \mathbb{E}_{tr \in T_c} [f(\mathbf{tr})] + b \times \text{Std}_{tr \in T_c} [f(\mathbf{tr})] \quad (8)$$

In this equation, T_c is the set of triplets whose relation is in the causal relation set R_c , $f(\mathbf{tr})$ represents the score of the triplet tr , $b \in \mathbb{R}$ is a constant, $\mathbb{E}$ denotes the expectation, and Std represents the standard deviation.

Remark 2: For large-scale KGs with extensive knowledge, it is unnecessary to complete the KG, which can be time-consuming. In such cases, we directly extract triplets with causal relations present in the KG as causal knowledge:

$$S = \{(h, r, t) \mid h, t \in E_V, r \in R_c, (h, r, t) \in T\}$$

For the triplets in S , we identify ancestral constraints for structure learning as follows:

$$\Pi = \{(X_i, X_j) \mid X_i = M^{-1}(h), X_j = M^{-1}(t), (h, r, t) \in S\} \quad (9)$$

Here, (X_i, X_j) indicates that at least one directed path exists in the causal structure G_T , and $M^{-1}(h)$ represents the variables corresponding to entity h . These ancestral constraints Π are then utilized in the structure learning process to guide the recovery of the causal DAG.

B. KG-Driven Structure Learning With Soft Scoring

To incorporate KG-derived prior constraints Π into the structure learning process, we employ a soft integration approach that extends the data approximation scoring function $\sigma(G; D)$. This scoring function is defined based on the log-likelihood of the data D given a causal structure G :

$$\sigma(G; D) = \log P(D \mid G) + \log P(G) \quad (10)$$

Typically, $P(G)$ is assumed to be uniform across all possible structures G in the absence of prior knowledge.

To integrate the priors Π , we modify the scoring function by incorporating the conditional probability $P(G \mid \Pi)$:

$$\sigma_p(G; D, \Pi) = \log P(D \mid G) + \log P(G \mid \Pi) \quad (11)$$

To compute $\log P(G \mid \Pi)$, we introduce a hyper-parameter $\lambda \in (0.5, 1)$ that represents the probability of the existence of a directed path $(X_i, X_j) \in \Pi$. Additionally, we approximate the joint probability of multiple path existences by factorizing the probability of individual paths, leading to the following formulation:

$$\log P(G \mid \Pi) = \sum_{(X_i, X_j) \in \Pi} \log (\lambda \mathbf{1}(X_i \rightsquigarrow X_j) + (1 - \lambda) \mathbf{1}(X_i \not\rightsquigarrow X_j)) \quad (12)$$

where $\mathbf{1}(\cdot)$ is the indicator function that takes the value 1 if the condition inside holds and 0 otherwise. Here, $X_i \rightsquigarrow X_j$ denotes

Algorithm 1: Soft KG-Driven Structure Learning.

Input: Knowledge Graph K ; Descriptions of Variables X ; Observational Data D

Output: DAG G

- 1 $\Pi \leftarrow$ Derive ancestral constraints among X from K
// Follow Section III-A
- 2 $\lambda \leftarrow$ pre-set prior probability
- 3 Define the prior-based scoring criteria
 $\sigma_p(G; D, \Pi) := \sigma_{MML}(G; D) + b(G; \Pi)$
// $\sigma_{MML}(G; D)$ is the MML scoring function [55]; $b(G; \Pi)$ is defined by Equation (12)
- 4 $W \leftarrow$ Initialize the weight of DAGs with 0
- 5 $O, G \leftarrow$ Find an initial solution of ordering and structure by Simulated Annealing
- 6 **for** $i = 1$ **to** L **do**
// L is the pre-set iteration number
 - 7 $O', G' \leftarrow$ Randomize a neighbor of O, G
 - 8 **if** $\frac{\sigma_p(G'; D, \Pi) - \sigma_p(G; D, \Pi)}{T} > \log U[0, 1]$ **then**
 - 9 $G'_{clean} \leftarrow$ delete edges from G' whenever this reduces $\sigma_p(G'; D, \Pi)$
 - 10 $W[G'_{clean}] \leftarrow$
 $W[G'_{clean}] + \frac{\exp(\sigma_p(N_{G0}; D, \Pi)) - \exp(\sigma_0)}{T}$
 - 11 $O, G \leftarrow O', G'$
- 12 $W' \leftarrow$ Group DAGs in W using KL divergence test and accumulate the corresponding weights
- 13 $Gr \leftarrow$ The group in W' with highest weight
- 14 $G \leftarrow$ The DAG with the highest weight in Gr
- 15 **return** G

the existence of a directed path from X_i to X_j in G , while $X_i \not\rightarrow X_j$ indicates the absence of such a path.

Remark 3: The existence of multiple paths is not inherently independent, making the calculation of joint probabilities complex. However, we approximate the joint probability by factorizing it into the product of marginal probabilities for individual paths. This simplification is justified as the underlying graph is typically sparse, and the presence of one path does not significantly influence the probability of another.

Denoting $\log P(G | \Pi)$ as $b(G; \Pi)$, the prior-based scoring function is formulated as:

$$\sigma_p(G; D, \Pi) = \sigma(G; D) + b(G; \Pi) \quad (13)$$

where $\sigma(G; D)$ is the original scoring function that approximates the log-likelihood of the data $\log P(D | G)$, and $b(G; \Pi)$ represents the prior adherence term defined in (12).

For the search process, we utilize the method proposed by O'Donnell et al. [55], which employs a Markov Chain Monte Carlo (MCMC) search as the backbone algorithm. The pseudocode for our soft KG-driven causal structure learning framework is presented in Algorithm 1.

C. Selection of Prior Probability λ

To establish criteria for selecting the hyper-parameter $\lambda \in (0.5, 1)$, we first examine its influence on the scoring of DAGs. Specifically, consider the bonus score $b(\cdot)$ that rewards adherence to a single constraint, which is calculated as:

$$\hat{b} = \log \lambda - \log(1 - \lambda) \quad (14)$$

We parameterize λ as $1 - 10^{-k}$, where $k \in \mathbb{N}^+$. Substituting this into the bonus score, we obtain:

$$\hat{b} = \log(10^k - 1) \approx k \log 10 \quad (15)$$

This approximation indicates that the bonus score scales linearly with k . Therefore, by incrementally increasing k , we can effectively adjust the influence of λ on structure learning.

For selecting the optimal k , we recommend starting with a relatively small value (e.g., $k = 1$, corresponding to $\lambda = 0.9$) and incrementally increasing k . For each k , we observe the count $c(k)$ of prior constraints that are satisfied in the resulting DAG. We select k such that:

$$c(k+1) - c(k) < \frac{1}{2} (c(k) - c(k-1)) \quad (16)$$

This criterion assumes that the correct priors exhibit a relatively low degree of conflict with the data approximation, and identifies a threshold where the most accurate priors are accepted, while erroneous priors are excluded.

Remark 4: It is important to note that there is no theoretical guarantee that correct priors will always exhibit less conflict with data approximation compared to erroneous ones. However, our empirical results (see Fig. 4) demonstrate that the priors accepted under smaller values of λ tend to be more accurate than those that are not.

D. Alternative Approaches for KG-Driven Structure Learning

This part presents alternative approaches to the soft score-based method, aiming to enhance the utility of our framework. These alternatives include a hard score-based approach that ensures strict adherence to all priors, a constraint-based approach, and a gradient-based approach tailored to integrate with other mainstream DAG structure learning algorithms.

Score-Based Approach with Hard Constraints: For comparison, we also implement a hard integration method for KG-derived priors Π , ensuring that all constraints are strictly satisfied. In this context, each prior constraint is assigned a probability of 1. According to (12), any violation of a constraint imposes a penalty of negative infinity on the score, resulting in the following scoring function:

$$\sigma_H(G; D, \Pi) = \sigma(G; D) - \infty \times \psi(G; \Pi) \quad (17)$$

Here, $\psi(G; \Pi)$ denotes the number of ancestral constraints satisfied within the structure G , and ∞ represents a sufficiently large positive value to ensure that only structures fully satisfying Π are considered optimal. To optimize the hard prior constraint score σ_H , we adopt the ordering-based search strategy proposed by Li et al. [56].

Constraint-Based Approach with Constraints Reduced to Edge Absence: In constraint-based approaches, such as the PC

algorithm [57], the structure is derived based on conditional independence (CI) tests. However, integrating ancestral constraints directly into these algorithms is typically ineffective. For instance, the PC algorithm starts with a fully connected graph and removes edges that are conditionally independent given certain variable sets. Since the algorithm lacks a mechanism for adding edges, any edges removed from true paths specified by Π cannot be recovered, preventing the correct identification of constrained paths.

To overcome this limitation, we transform ancestral constraints Π into a weaker form that enforces the absence of edges. For a directed path $(X_i, X_j) \in \Pi$, acyclicity ensures that the reverse edge (X_j, X_i) must not be present in the graph. Given the transitive nature of directed paths, i.e., if (X_i, X_j) and (X_j, X_k) are present, then (X_i, X_k) is implied, we derive the following edge absence constraints:

$$(X_j, X_i) \notin E(G) \quad \forall (X_i, X_j) \in \Pi^+ \quad (18)$$

where Π^+ is the transitive closure of Π , containing all paths derivable from Π . It is formally defined as:

Definition 1 (Transitive Closure): The transitive closure of a set R of ordered pairs is the smallest relation R^+ on the same set that contains R and is transitive. Formally, R^+ includes all pairs from R , and for any elements a, b , and c , if $(a, b) \in R^+$ and $(b, c) \in R^+$, then $(a, c) \in R^+$.

In the PC algorithm, which begins with a complete graph G_c where $E(G_c) = \{(X_i, X_j) \mid i \neq j\}$, we remove the edges as defined by (18). We then apply the PC algorithm on the modified graph G_c^Π to derive the final structure:

$$G \leftarrow \text{PC}(D, G_c^\Pi) \text{ where} \\ E(G_c^\Pi) = E(G_c) \setminus \{(X_i, X_j) \mid (X_j, X_i) \in \Pi^+\} \quad (19)$$

This edge absence constraint effectively guides the structural learning process based on CI tests, albeit with reduced prior information.

Gradient-Based Approach with Partial Order Constraints: The gradient-based approach [58] formulates the structure learning task as a continuous optimization problem with an equality constraint enforcing acyclicity:

$$\min_{W \in \mathbb{R}^{d \times d}} F(W; D) \quad \text{s.t.} \quad h(W) = 0 \quad (20)$$

Here, W is the adjacency matrix where $W_{i,j} \neq 0$ indicates a directed edge from X_i to X_j . $F(W; D)$ represents the loss function for data approximation, and $h(W) = \text{Trace}(e^{W \circ W}) - d$ ensures that the graph is acyclic. This optimization problem is typically solved using gradient-based methods, which can struggle to recover correct edges of directed paths in ancestral constraints due to the propensity of these methods to converge to local optima that may introduce erroneous edges.

To mitigate this issue, we employ KG-derived causal knowledge as partial order constraints, which are a weaker form of ancestral constraints. Partial order constraints can be transformed into a set of path-forbidden constraints, thereby preventing the recovery of erroneous edges that are not included in the specified paths. Moreover, partial order constraints are more robust

than ancestral constraints, maintaining correctness for $X_i \prec X_j$ provided no directed path exists from X_i to X_j .

Formally, we define Π as the set of partial order constraints, where $(X_i, X_j) \in \Pi$ constrains that X_i precedes X_j in the ordering, denoted as $X_i \prec X_j$. This constraint ensures that there is no directed path from X_j to X_i in the graph. Considering the transitive nature of partial orders, we have the following result:

$$G \text{ satisfies } \Pi \iff \text{no path exists from } X_j \text{ to } X_i \\ \text{in } G \quad \forall (X_i, X_j) \in \Pi^+ \quad (21)$$

where Π^+ is the transitive closure of Π . The constraint on the right-hand side of (21) can be formulated as:

$$O(W, \Pi) \equiv \sum \mathbf{M} \left(\sum_{k=1}^d W_{\Pi}^k \right) \circ \left(\sum_{k=1}^d (W \circ W)^k \right) = 0 \quad (22)$$

where $W_{\Pi} \in \{0, 1\}^{d \times d}$ is the mask of Π defined by $(W_{\Pi})_{i,j} = 1 \iff (X_i, X_j) \in \Pi$. The function $\mathbf{M}(\cdot)$ transforms the inner matrix into a binary mask, where $(\mathbf{M}(A))_{i,j} = 1$ if $A_{i,j} \neq 0$ and 0 otherwise.

Under this continuous equality constraint derived from the KG-derived prior Π , the optimization task is formulated as:

$$\min_{W \in \mathbb{R}^{d \times d}} F(W; D) \quad \text{s.t.} \quad h(W) + O(W, \Pi) = 0 \quad (23)$$

We employ the NOTEARS method [58] as the backbone algorithm, where $F(W; D) = \|DW - D\|_2^l + \lambda_1 |W|$, and $h(W) = \text{Trace}(e^{W \circ W}) - d$. The augmented Lagrangian method is used to solve this constrained optimization problem.

E. Analysis of Time Complexity

To analyze the time complexity of our method, we assume there are d nodes and m samples, with a sparse graph containing $e = O(d)$ edges, and the size of the ancestral constraints set Π is a .

For the score-based approach, we extend the scoring function by adding a term that verifies the existence of each path in Π . The existence of a path can be checked using a graph traversal algorithm like BFS or DFS starting from the source node, which takes $O(d + e) = O(d)$ time. Thus, verifying all paths in Π requires $O(ad)$ time. The BIC score, used as the scoring function for estimation, has a time complexity of approximately $O(md)$ for sparse graphs. Let S_d represent the time complexity of the backbone search algorithm. The total time complexity of our approach is $O(S_d(a + m)d)$, which remains comparable to the original $O(S_dmd)$, as the number of priors a is usually smaller than the sample size m .

For constraint-based approaches, the time complexity remains unchanged from the data-based counterpart since we only remove edges from the initial graph. For gradient-based approaches, the time complexity increases by $O(d^4)$ per training step due to the calculation of (22).

TABLE I
THE DETAILS OF CAUSAL STRUCTURE DATASETS

Dataset	@V	@E	@P	Research topic	Sample size
Asia	8	8	18	Reasons causing dyspnoea	250 / 1000
Child	20	25	230	Childhood diseases	500 / 2000
Alarm	37	46	509	ALARM monitoring system	1000 / 4000
Insurance	27	52	1008	Car insurance cost	500 / 2000
Mildew	35	46	540k	Winter wheat mildew control	8000 / 32000
Win95pts	76	112	574	Windows performance testing	8000 / 32000

@V, @E, @P represent the number of variables, edges and parameters of the BNs, respectively.

IV. EXPERIMENTS

A. Datasets and Setup

We utilize three knowledge graphs (KGs) for prior derivation. The first KG, constructed by Daowd et al. [59], focuses on chronic diseases and cancers and comprises approximately 70,000 entities and 200,000 triplets, which we use for the medical domain. The second KG is Wikidata [60], a large-scale, cross-domain KG containing over 100 million entities and 1.5 billion triplets. The third KG is ConceptNet [61], containing 8 million entities and 38 million triplets. Wikidata and ConceptNet are employed for other domains. Due to their convenient retrieval interfaces, we directly use these interfaces for entity-to-variable matching and causal triplet extraction.

For causal structure learning, we employ six datasets from the Bayesian Network Repository,² selecting three from the medical domain to correspond with the domain-specific KG and three from other diverse domains to align with the cross-domain KG. For each dataset, we simulate observational data with two different sample sizes based on the parameters of the BN, conducting six repetitions for each sample size. Detailed information about the datasets is provided in Table I.

We investigate various data-based structure learning algorithms spanning mainstream approaches. For score-based methods, we employ a Tabu search algorithm with the Bayesian Dirichlet equivalent uniform (BDeu) scoring function [56] for the hard approach and a MCMC search method with the Minimum Message Length (MML) scoring function [62] for the soft approach. Additionally, we utilize the PC algorithm [57] for the constraint-based method and the NOTEARS algorithm [58] for the gradient-based method. The specifics of the algorithms used are summarized in Table II.

For text embedding, we employ the Sentence-BERT model, which is pre-trained on a large corpus.³ For the KG embedding process, we use Amazon's DGL-KE KG embedding framework⁴ to train the TransE model.

As for hyperparameter setting, the threshold α in (4) is set to 0.85, b in (7) is set to 0.5, and the prior probability λ in the soft approach is set to 0.999 as default.

All experiments are run on an AMD Ryzen9 7950x CPU @ 4.5 GHz with a memory limit of 32 GB and a GeForce RTX 3090 GPU with 24 GB graphics memory.

²<https://www.bnlearn.com/bnrepository/>

³Available at <https://github.com/UKPLab/sentence-transformers> [63].

⁴<https://github.com/awsmlabs/dgl-ke/>

TABLE II
UTILIZED STRUCTURE LEARNING APPROACHES ACROSS VARIOUS RESEARCH LINES IN THE EXPERIMENT

Method	Type	Model	Prior	Way
MCMC	Score-based	BN	-	-
TABU	Score-based	BN	-	-
PC	Constraint-based	BN	-	-
NOTEARS	Gradient-based	SEM	-	-
KG-MCMC	Score-based	BN	Path Existence	Soft
KG-TABU	Score-based	BN	Path Existence	Hard
KG-PC	Constraint-based	BN	Post Hoc	-
KG-NOTEARS	Gradient-based	SEM	Partial Order	Hard

TABLE III
DETAILS OF KG-DERIVED PRIORS FROM WIKIDATA ON THREE DATASETS

Dataset	@V	KG-derived Prior Knowledge		
		@V Covered	@Correct	@Error
Insurance	27	19	15	1
Mildew	35	35	19	0
Win95pts	76	58	24	2

'@V Covered' denotes the number of variables included in the KG, '@Correct' represents the number of correct priors and '@Error' represents the number of erroneous ones.

B. Metrics

We use two structural evaluation metrics, the Structural Hamming Distance (SHD) and the F1 score.

SHD is the most popular structural metric, representing the count of different edges between the evaluated structure and the truth, and is smaller to be better. However, using solely this metric can not always properly represent the quality of the output structure. For example, an empty graph has an SHD value of the number of edges, which can be even better than the output of state-of-the-art algorithms in some cases.

To provide a more comprehensive evaluation, the F1 score is used, which is calculated by the harmonic mean of precision and recall rate of the recovered edges and provides a more balanced evaluation of the quality of structure learning.

C. Analysis of the KG-Derived Prior

We compare the causal triplets extracted from the KG with the true causal structures. The results for the medical domain datasets are illustrated in Fig. 2, while those for the cross-domain datasets are presented in Table III.

Our analysis reveals that the extracted causal knowledge predominantly corresponds to indirect causal paths within the true causal structures. This observation validates our approach of employing ancestral constraints for integrating KG-derived causal knowledge into the structure learning process. Furthermore, the majority of KG-derived priors align correctly with the paths in the true structure, underscoring the practical effectiveness of our KG-driven causal discovery method.

However, some inaccuracies are present, which appear to stem from the inherent complexity of accurately representing genuine causal interactions within knowledge bases. For instance, the KG-derived causal relationship between HypDistrib (a

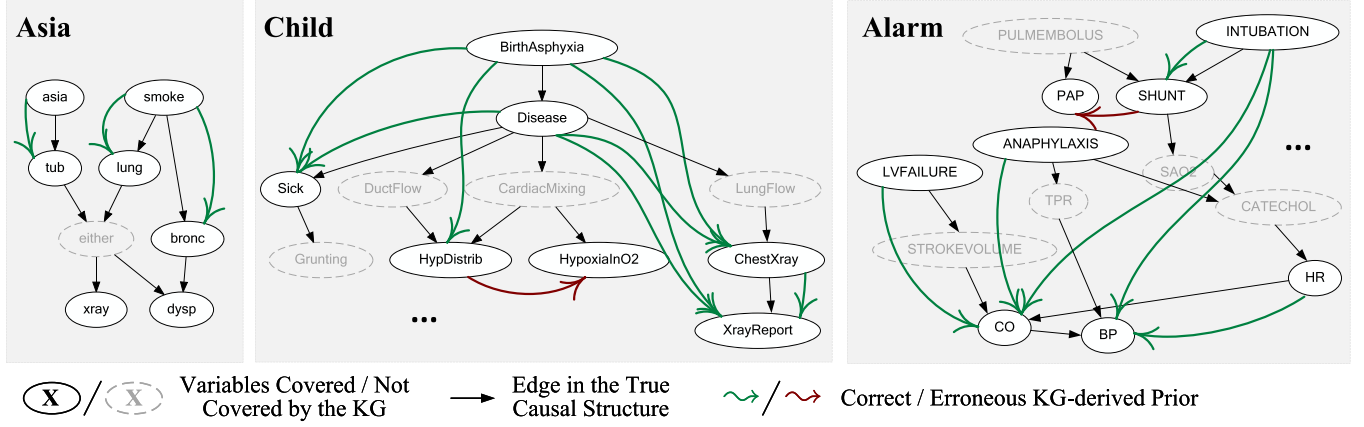


Fig. 2. Visualization of KG-derived Priors on the Three Datasets. The structure of *Asia* dataset is completely presented, while the other datasets' structures are presented with the parts that are related to the KG-derived prior.

TABLE IV
COMPARATIVE PERFORMANCE OF SCORE-BASED STRUCTURE LEARNING WITH THE KG-DRIVEN METHOD

Dataset	Size	Performance of Data-based / KG-driven CSL							
		MCMC / KG-MCMC (Soft)				TABU / KG-TABU (Hard)			
		SHD↓	Δ↓	F1↑	Δ↑	SHD↓	Δ↓	F1↑	Δ↑
Asia	250	3.7 / 0.8	-77.3%	0.67 / 0.95	+42.1%	4.2 / 2.5	-40.0%	0.68 / 0.83	+21.7%
	1000	2.7 / 0.5	-81.3%	0.74 / 0.97	+30.8%	2.5 / 1.5	-40.0%	0.75 / 0.88	+17.0%
Child	500	6.7 / 6.0	-10.0%	0.81 / 0.83	+10.3%	9.5 / 9.8	+3.5%	0.72 / 0.72	-0.3%
	2000	2.0 / 1.0	-50.0%	0.92 / 0.96	+4.3%	5.3 / 3.8	-28.1%	0.82 / 0.91	+10.8%
Alarm	1000	11.7 / 11.3	-2.9%	0.82 / 0.82	+0.4%	9.5 / 9.8	+3.5%	0.88 / 0.88	-0.4%
	4000	9.0 / 8.3	-7.4%	0.85 / 0.87	+1.4%	6.5 / 8.0	+23.1%	0.91 / 0.89	-2.4%
Insurance	500	35.3 / 28.7	-18.9%	0.47 / 0.62	+32.8%	25.7 / 26.3	+2.6%	0.68 / 0.69	+1.2%
	2000	34.8 / 27.3	-21.5%	0.49 / 0.62	+25.3%	15.0 / 18.0	+20.0%	0.81 / 0.79	-2.1%
Mildew	8000	48.5 / 48.2	-0.7%	0.26 / 0.28	+9.2%	22.8 / 22.0	-3.6%	0.63 / 0.68	+7.1%
	32000	63.3 / 57.3	-9.5%	0.19 / 0.28	+47.1%	21.0 / 13.0	-38.1%	0.67 / 0.79	+17.3%
Win95pts	8000	63.4 / 63.2	-0.3%	0.68 / 0.70	+2.4%	FNR / FNR	FNR	FNR / FNR	FNR
	32000	79.6 / 55.7	-30.1%	0.62 / 0.68	+8.9%	FNR / FNR	FNR	FNR / FNR	FNR

The column labeled Δ quantifies the relative performance improvement of KG-driven over the Data-based structure learning. Entries reflecting performance enhancements are marked in red, while diminished performance instances are indicated in blue.

hypothetical condition related to blood or oxygen distribution within the body) and *HypoxiaInO2* (a state characterized by insufficient oxygen reaching body tissues) is supported by medical literature [64], where issues in blood distribution can lead to inadequate oxygen supply. Nevertheless, the actual interaction between these entities manifests as a common-cause structure rather than a unidirectional causal flow. This discrepancy highlights the necessity for robust error handling mechanisms in the structure learning process to mitigate the impact of such inaccuracies.

D. Comparison Between KG-Driven and Data-Based Causal Discovery

We compare our KG-driven causal discovery approach with its data-based counterpart. Specifically, the score-based and constraint-based methods are applied to data generated from Bayesian Network (BN) models, while the gradient-based approach is applied to data generated from Structural Equation Models (SEMs) with linear functions and Gaussian noise.

TABLE V
COMPARATIVE PERFORMANCE OF CONSTRAINT-BASED STRUCTURE LEARNING WITH THE KG-DRIVEN METHOD

Dataset	Size	Performance of PC / KG-PC			
		SHD↓	Δ↓	F1↑	Δ↑
Asia	250	1.3 / 1.3	0.0%	0.62 / 0.69	+11.3%
	1000	2.2 / 2.3	+7.7%	0.57 / 0.62	+9.2%
Child	500	32.5 / 32.8	+1.0%	0.28 / 0.29	+2.4%
	2000	35.8 / 36.2	+0.9%	0.30 / 0.30	+1.8%
Alarm	1000	66.8 / 67.7	+1.2%	0.25 / 0.25	+0.3%
	4000	84.0 / 86.0	+2.4%	0.23 / 0.23	-0.2%
Insurance	500	64.7 / 65.8	+1.8%	0.22 / 0.23	+2.0%
	2000	63.3 / 64.0	+1.1%	0.27 / 0.28	+3.9%
Mildew	8000	71.2 / 70.7	-0.7%	0.29 / 0.30	+3.4%
	32000	63.0 / 63.8	+1.3%	0.33 / 0.33	+1.4%
Win95pts	8000	107 / 109	+1.6%	0.36 / 0.37	+3.0%
	32000	92.0 / 94.0	+2.2%	0.40 / 0.41	+2.2%

Results on the datasets are evaluated using Structural Hamming Distance (SHD) and F1 score, and are reported in Table IV for score-based methods, Table V for the constraint-based method, and Table VI for the gradient-based method. Additionally, run time results are presented in Fig. 3.

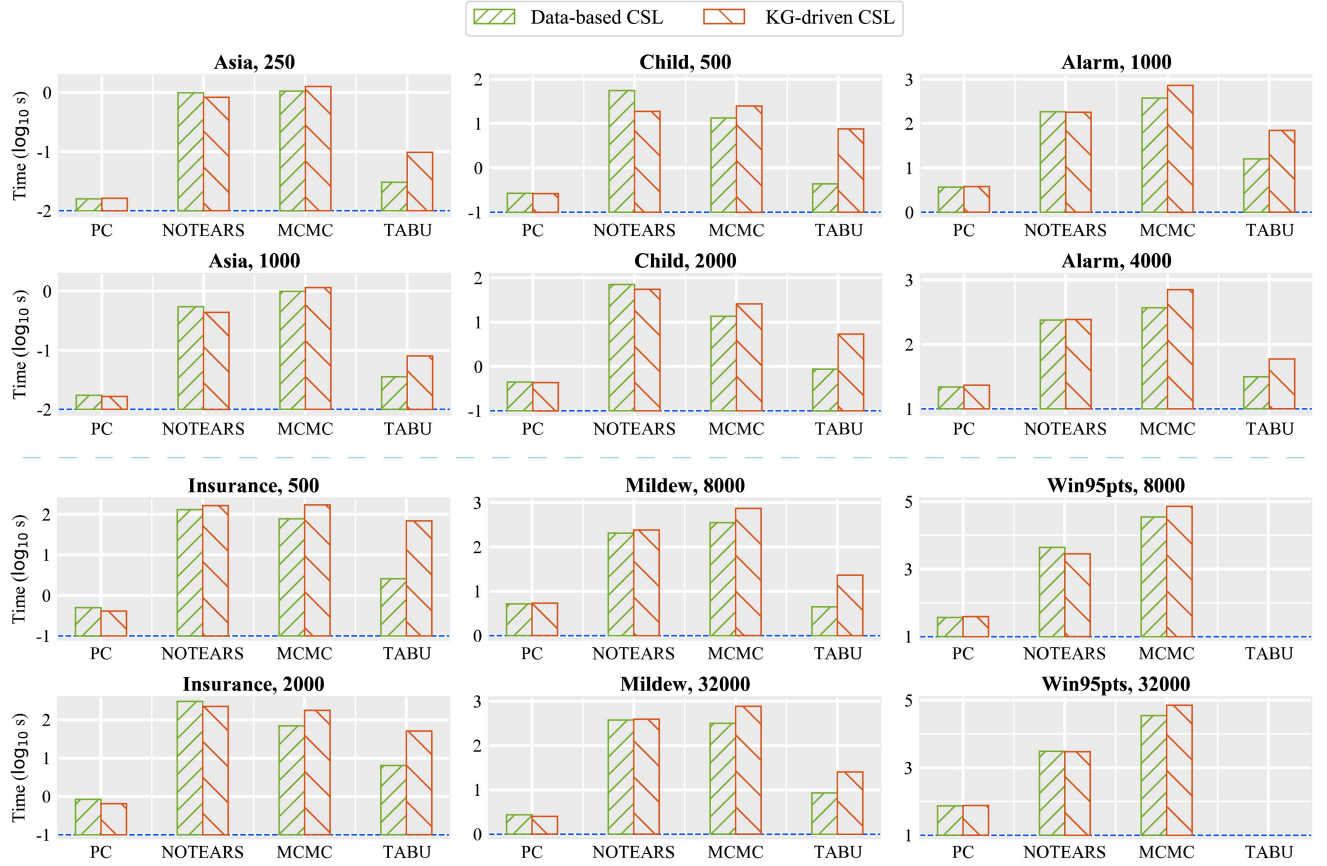


Fig. 3. Comparative Run Time ($\log s$) between Data-based and KG-driven structure learning.

TABLE VI
COMPARATIVE PERFORMANCE OF GRADIENT-BASED STRUCTURE LEARNING
WITH THE KG-DRIVEN METHOD

Dataset	Size	Performance of NOTEARS / KG-NOTEARS			
		SHD↓	Δ ↓	F1↑	Δ ↑
Asia	20	2.2 / 1.5	-30.8%	0.86 / 0.91	+6.0%
	1000	0.0 / 0.0	0.0%	1.0 / 1.0	0.0%
Child	20	19.0 / 18.7	-1.8%	0.57 / 0.57	+0.8%
	1000	7.5 / 3.8	-48.9%	0.83 / 0.91	+10.5%
Alarm	20	45.7 / 42.8	-6.2%	0.45 / 0.47	+6.0%
	1000	12.5 / 7.5	-40.0%	0.85 / 0.91	+6.8%
Insurance	20	46.2 / 39.5	-14.4%	0.44 / 0.55	+22.6%
	1000	44.0 / 27.0	-31.6%	0.45 / 0.69	+26.3%
Mildew	20	45.5 / 42.2	-7.3%	0.44 / 0.50	+14.8%
	1000	28.8 / 15.8	-45.1%	0.64 / 0.81	+26.4%
Win95pts	20	108 / 97.0	-9.9%	0.40 / 0.47	+15.7%
	1000	20.8 / 18.0	-13.6%	0.90 / 0.92	+1.5%

Our findings indicate that for score-based methods, the soft approach consistently outperforms the data-based approach, whereas the hard approach underperforms its data-based counterpart on datasets with erroneous KG-derived priors. This demonstrates the robustness of the soft approach in KG-driven structure learning. For the constraint-based approach, we observe an improvement in F1 score but a degradation in SHD. This occurs because the edge absence constraints force the

recovery of additional edges during the CI tests, increasing the recall of true edges at the cost of introducing erroneous ones. In the case of gradient-based methods, the KG-driven approach outperforms the data-based counterpart. This is attributed to the use of partial order constraints, which are less restrictive but more robust in scenarios where variables do not have direct ancestor-descendant relationships.

Regarding time efficiency, we find that KG-driven approaches are slower than data-based methods due to the additional complexity introduced by integrating structural constraints. Nevertheless, the difference in run time is not substantial, remaining within acceptable limits for practical applications.

E. Impact of Prior Quality on KG-Driven CSL

To evaluate the robustness of KG-driven causal structure learning (CSL), particularly the soft constraint approach, we conducted simulations with ancestral constraints of varying accuracy across three datasets. Specifically, we selected 30% of variable pairs with directed paths as correct priors and introduced erroneous constraints at ratios of 0 (error-free), 0.2, and 1 (massive errors). For the soft constraint approach, the prior probability was set to 0.9. The results comparing prior-based CSL to data-driven CSL are presented in Table VII.

TABLE VII
PERFORMANCE COMPARISON BETWEEN THE SOFT AND HARD KG-DRIVEN CSL WITH VARIOUS QUALITIES OF PRIORS

Dataset	Size	Ratio	Performance of KG-MCMC (Soft) / KG-TABU (Hard)			
			SHD↓	Δ ↓	F1↑	Δ ↑
Asia	250	0	1.5±1.4 / 3.3±3.6	-59% / -21%	0.89±0.10 / 0.77±0.24	+33% / +13%
		0.2	1.3±1.2 / 4.5±4.8	-64% / +7%	0.91±0.09 / 0.72±0.32	+35% / +5%
		1	4.5±1.4 / 5.3±3.1	+22% / +26%	0.67±0.10 / 0.65±0.20	0% / -4%
	1000	0	0.5±0.5 / 0.5±0.5	-81% / -80%	0.97±0.04 / 0.96±0.05	+31% / +28%
		0.2	0.3±0.5 / 1.8±0.8	-88% / -27%	0.98±0.03 / 0.86±0.09	+32% / +15%
		1	3.3±0.5 / 4.3±0.5	+23% / +78%	0.73±0.02 / 0.70±0.02	-2% / -7%
Child	500	0	3.5±2.3 / 8.7±1.0	-48% / -9%	0.91±0.05 / 0.78±0.03	+12% / +8%
		0.2	3.3±1.9 / 12.5±1.5	-50% / +32%	0.91±0.04 / 0.70±0.04	+12% / -3%
		1	6.7±1.6 / 14.8±2.5	0% / +56%	0.83±0.04 / 0.69±0.05	+2% / -5%
	2000	0	1.0±0.0 / 1.7±1.5	-50% / -69%	0.96±0.00 / 0.95±0.04	+4% / +16%
		0.2	1.0±0.0 / 6.5±0.8	-50% / +23%	0.96±0.00 / 0.86±0.02	+4% / +4%
		1	1.3±0.5 / 12.0±1.8	-33% / +126%	0.95±0.02 / 0.76±0.04	+3% / -8%
Alarm	1000	0	12.0±7.0 / 8.5±1.6	+3% / -11%	0.85±0.09 / 0.91±0.02	+4% / +3%
		0.2	16.5±3.3 / 21.7±1.5	+41% / +128%	0.80±0.05 / 0.75±0.02	-3% / -14%
		1	23.2±4.0 / 44.5±0.8	+98% / +368%	0.72±0.07 / 0.58±0.01	-12% / -34%
	4000	0	6.8±4.7 / 6.3±1.8	-24% / -3%	0.92±0.06 / 0.93±0.02	+8% / +2%
		0.2	14.2±2.4 / 19.3±2.2	+57% / +197%	0.83±0.03 / 0.78±0.03	-2% / -15%
		1	21.0±6.2 / 44.7±1.5	+133% / +587%	0.74±0.09 / 0.58±0.01	-13% / -37%

The ‘Ratio’ column specifies the proportion of erroneous to correct priors. The symbol Δ represents the relative performance improvement of the KG-driven method compared to its data-driven counterpart. Entries showcasing enhancement in performance relative to the data-driven method are emphasized in red, while diminished performance instances are indicated in blue.

Our analysis reveals that the soft approach generally offers greater performance enhancements or minimal degradation compared to the hard approach. Notably, with entirely accurate priors in the *Child* and *Alarm* datasets, the hard approach outperforms the soft approach by fully utilizing all correct constraints. However, as the proportion of erroneous priors increases, the hard constraint approach’s performance degrades sharply, falling below that of the data-driven method due to its inability to disregard incorrect priors. In contrast, the soft constraint approach remains robust, even outperforming data-driven CSL in the *Child* dataset with an error-to-truth ratio of 1. This resilience stems from the soft approach’s ability to reject priors that significantly conflict with the data distribution by balancing data fit with prior probability.

However, the effectiveness of the soft approach is contingent on the appropriate setting of the hyper-parameter λ . An unsuitable λ can result in the rejection of correct priors or the acceptance of erroneous ones, thereby compromising CSL outcomes. Future experiments will further investigate the influence of prior probability settings on CSL performance in the presence of imperfect priors.

F. Evaluating the Influence of Prior Probability Settings

This section investigates how varying prior probability values impact the performance of the soft constraint approach, particularly when handling imperfect priors. As outlined in Section III-B, prior strength is proportional to $\log(1 - p)^{-1}$. We experimented with prior probabilities set at 0.6, 0.9, 0.99, 0.999, and higher, though probabilities above 0.99999 were excluded due to floating-point overflow errors. The selected range effectively captures key patterns and trends.

We simulated priors by selecting 30% of correct ancestral constraints and adding erroneous constraints at a 0.2 ratio relative to the correct ones. Our analysis evaluated the precision and recall of the accepted priors, defining a prior as accepted if its indicated path is present in the output. Fig. 4 illustrates the precision, recall of accepted priors, and the F1-score of the output structure.

Across all scenarios, the precision of accepted priors matches or exceeds the accuracy of the input priors, highlighting the soft approach’s ability to discard erroneous priors effectively. Lower prior probabilities enhance error resistance by reducing the influence of priors and increasing rejection rates. Conversely, higher prior probabilities improve recall but may decrease precision as more erroneous priors are accepted.

Regarding output quality, there is an initial improvement with increasing λ as correct priors are incorporated. However, beyond a certain point, performance declines due to the acceptance of more erroneous priors. This trend underscores the necessity of fine-tuning λ based on confidence in the prior knowledge to balance the acceptance of correct priors against the risk of incorporating erroneous ones.

G. Limitations

Despite the promising experimental results, our KG-driven causal discovery approach has several limitations that users should consider in practice.

First, our method is applicable when the investigated variables correspond to well-defined and meaningful concepts that are represented in the knowledge graph. For instance, in studies examining the relationship between gene sequences and proteins [65], it may be challenging to find appropriate priors within existing knowledge bases. Second, as demonstrated in Tables III, IV, and VII, the priors extracted from KGs can sometimes be incorrect

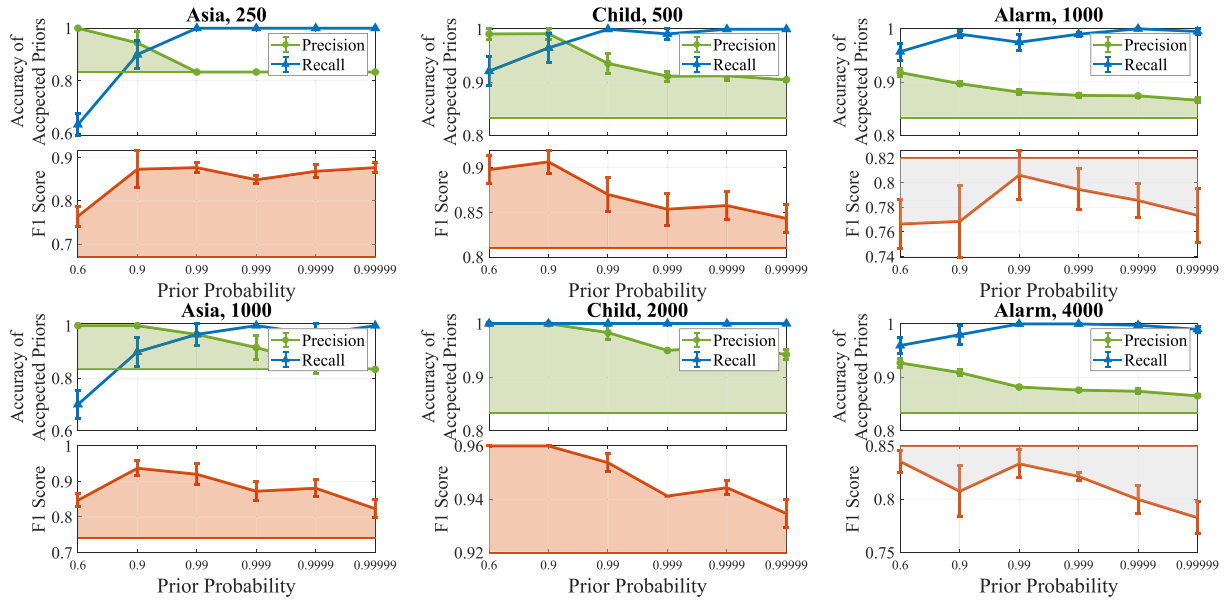


Fig. 4. Variation in accuracy of accepted priors and the F1 score for output structures generated by the soft KG-driven CSL across different prior probability settings and datasets. The green horizontal line represents the accuracy of the simulated priors at 0.83, while the orange horizontal line indicates the F1 score achieved by the data-driven algorithm on the same dataset. Areas where the soft constraint approach surpasses the baseline are marked in green (for prior precision) and orange (for F1 score), whereas regions where it does not are denoted in gray.

when applied as ancestral constraints. These inaccuracies may arise from errors in matching variables to entities or from the triplet extraction process, potentially leading to compromised structure recovery from the data.

To mitigate these issues, it is advisable to manually verify the KG-derived priors and eliminate those with ambiguous or unclear interaction mechanisms. This additional validation step can help ensure the reliability and accuracy of the recovered causal structures.

V. CONCLUSION

In this paper, we introduce a novel, expert-free framework for causal discovery that leverages prior causal knowledge extracted from knowledge graphs (KGs). By utilizing ancestral constraints derived from KGs, our approach integrates qualitative causal information into the causal structure learning (CSL) process. We employed a soft scoring method that incorporates these constraints into the scoring function, balancing adherence to prior knowledge with data-driven discovery and introducing tolerant ability to prior errors. Moreover, we also extend the utility of our framework across mainstream CSL algorithms and various application scenarios, including hard constraint integration, constraint-based methods and gradient-based methods. Our experimental results across various datasets and knowledge graphs indicate that the KG-driven approach enhances the accuracy of causal structure recovery compared to traditional data-driven methods.

Future work can focus on enhancing the effectiveness of the framework on high-dimensional data by incorporating accelerated structure learning strategies, such as the local-to-global approach introduced by [16]. Exploring the integration of KG-driven priors in discovery of dynamic causal relationships from

temporal data [17] and extending the approach to accommodate hidden variables or latent confounders are also promising directions.

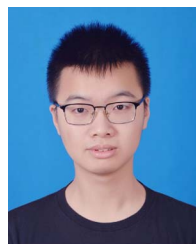
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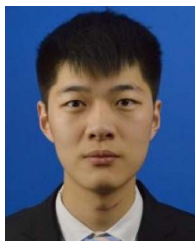
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